Executive Business Report

End-to-End Sales Forecasting & Demand Intelligence System

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Project: Internship Project – Week 3 & Week 4

Executive Summary

This project was developed to help retail businesses improve inventory planning by forecasting future sales, identifying unusual sales patterns, and analyzing product demand across different categories and regions. Historical Superstore sales data covering four years was analyzed using statistical and machine learning techniques. Multiple forecasting models were compared, anomalies in sales were detected, and products were segmented based on their demand behavior. An interactive Streamlit dashboard was also developed to present the results in a user-friendly manner for business decision-makers. The proposed system can support better stocking decisions, reduce inventory costs, and improve customer satisfaction.

Key Findings from Data Analysis & Forecasting

The exploratory data analysis revealed several important business insights.

- Technology generated the highest overall sales revenue among all product categories.
- Monthly sales showed a clear seasonal trend, with higher sales during festive and year-end periods.
- Sales generally increased over the four-year period, indicating positive business growth.
- Regional sales patterns varied, with some regions showing more consistent performance than others.
- Data quality was good with very few missing values and no significant duplicate records.

Three forecasting approaches were implemented and compared.

- SARIMA
- Prophet
- XGBoost

Among these models, **SARIMA** produced the most stable and reliable forecasts for the available monthly sales data and was selected as the recommended production model.

Three-Month Sales Forecast

Based on the selected forecasting model, the expected sales trend for the next three months indicates relatively stable business performance with normal seasonal fluctuations.

The forecast confidence intervals suggest that although small monthly variations are expected, no major decline in demand is anticipated during the forecast horizon. Therefore, inventory planning should remain close to the predicted demand while maintaining sufficient safety stock for seasonal variations.

Top Three Sales Anomalies

Two anomaly detection techniques (Isolation Forest and Z-Score) were used to identify unusual sales periods.

The most significant anomalies were observed during:

1. Year-End Holiday Season

Sales increased sharply due to holiday shopping and promotional campaigns.

2. Festival Sales Period

Temporary spikes in customer demand resulted in unusually high sales compared to normal weeks.

3. Low Sales Period

Certain weeks experienced lower-than-expected sales, possibly due to reduced customer activity or seasonal demand fluctuations.

Both anomaly detection methods identified similar abnormal periods, increasing confidence in the detected events.

Product Demand Segmentation

K-Means clustering grouped products into different demand categories based on sales volume, growth rate, volatility, and average order value.

The major demand segments include:

High Volume Stable Demand

Products with consistently strong sales that require continuous inventory availability.

Growing Demand

Products showing increasing customer interest and requiring gradual inventory expansion.

Low Volume High Volatility

Products with irregular demand that should be stocked cautiously to reduce holding costs.

Declining Demand

Products with decreasing sales that should be monitored carefully and reordered conservatively.

These demand segments provide valuable guidance for inventory planning and warehouse management.

Business Recommendations

Based on the complete analysis, the following recommendations are proposed:

Recommendation 1

Increase inventory levels before festive and holiday seasons since historical sales consistently show significant seasonal demand growth.

Recommendation 2

Maintain higher safety stock for Technology products because they generate the highest revenue and experience strong customer demand.

Recommendation 3

Use anomaly detection as an early warning system to identify unexpected sales spikes or drops, allowing management to respond quickly with inventory adjustments and marketing strategies.

Risk and Limitation

The forecasting models rely on historical sales patterns. Unexpected business events such as economic changes, supply chain disruptions, new competitors, or major promotional campaigns may significantly affect future sales and reduce forecast accuracy. Therefore, the forecasting model should be retrained periodically using the latest available sales data.

Conclusion

The developed Sales Forecasting and Demand Intelligence System successfully integrates time series forecasting, anomaly detection, demand segmentation, and interactive dashboard visualization into a single decision-support platform. The system enables data-driven inventory planning, improves forecasting accuracy, supports proactive business decisions, and provides valuable insights for retail managers. With regular model updates and continuous monitoring, the proposed solution can contribute to improved operational efficiency and better inventory management.